Build Your First AI Chatbot Using Python

Summary:-

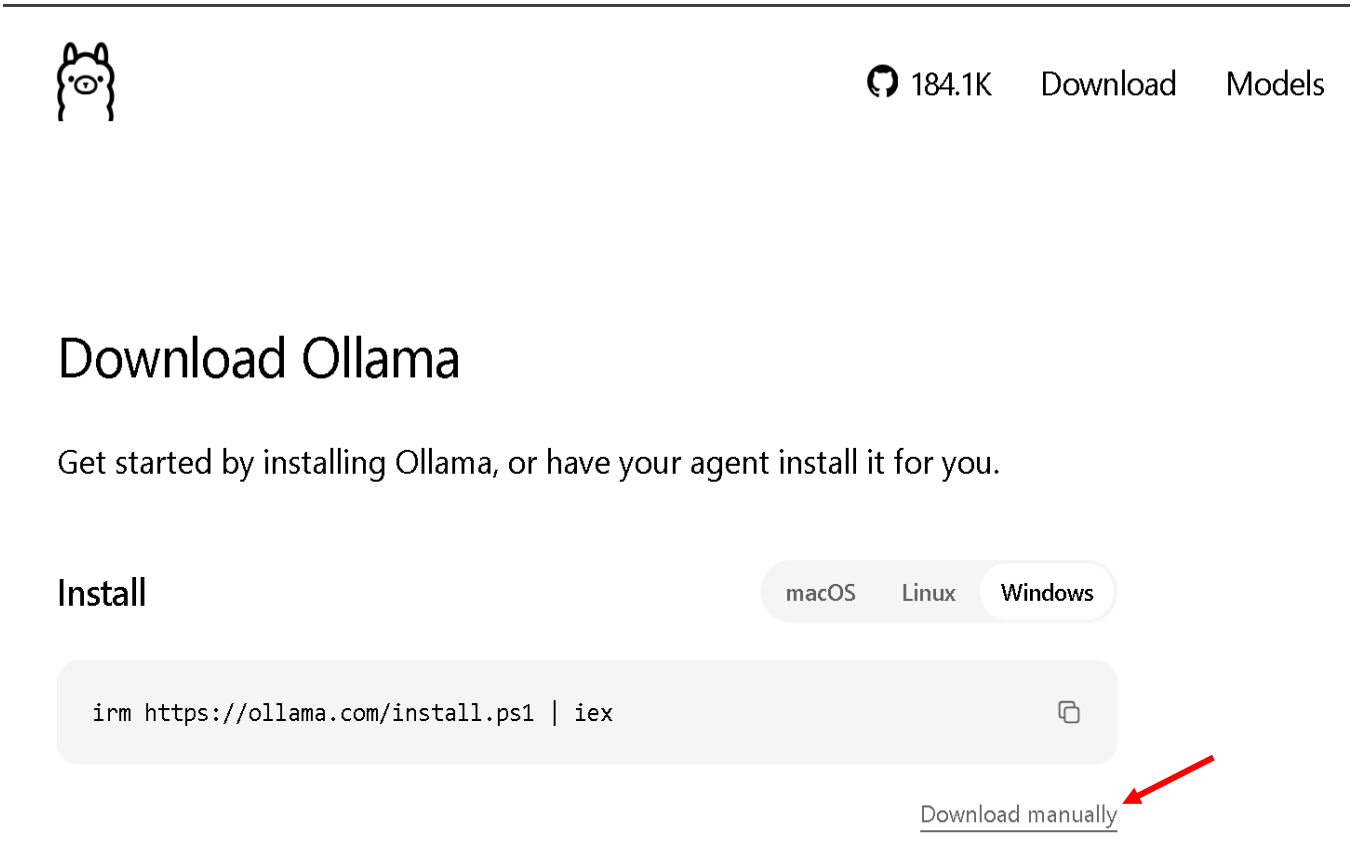
We created a basic AI chatbot using Python, Ollama, and the Qwen3:4B model. Using VS Code and the OpenAI Python library, we developed an interactive assistant that answers user questions through the terminal without requiring a paid cloud AI API.

To Build AI Applications we'll use free tools: -

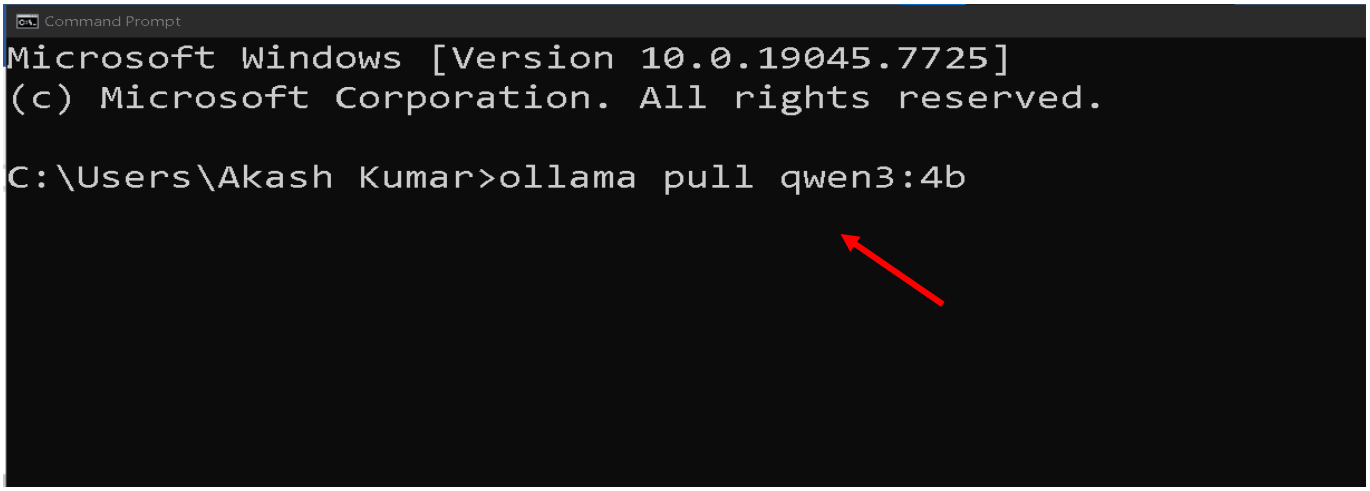
- Python
- Visual Studio Code
- Ollama
- Git

Lets Start: -

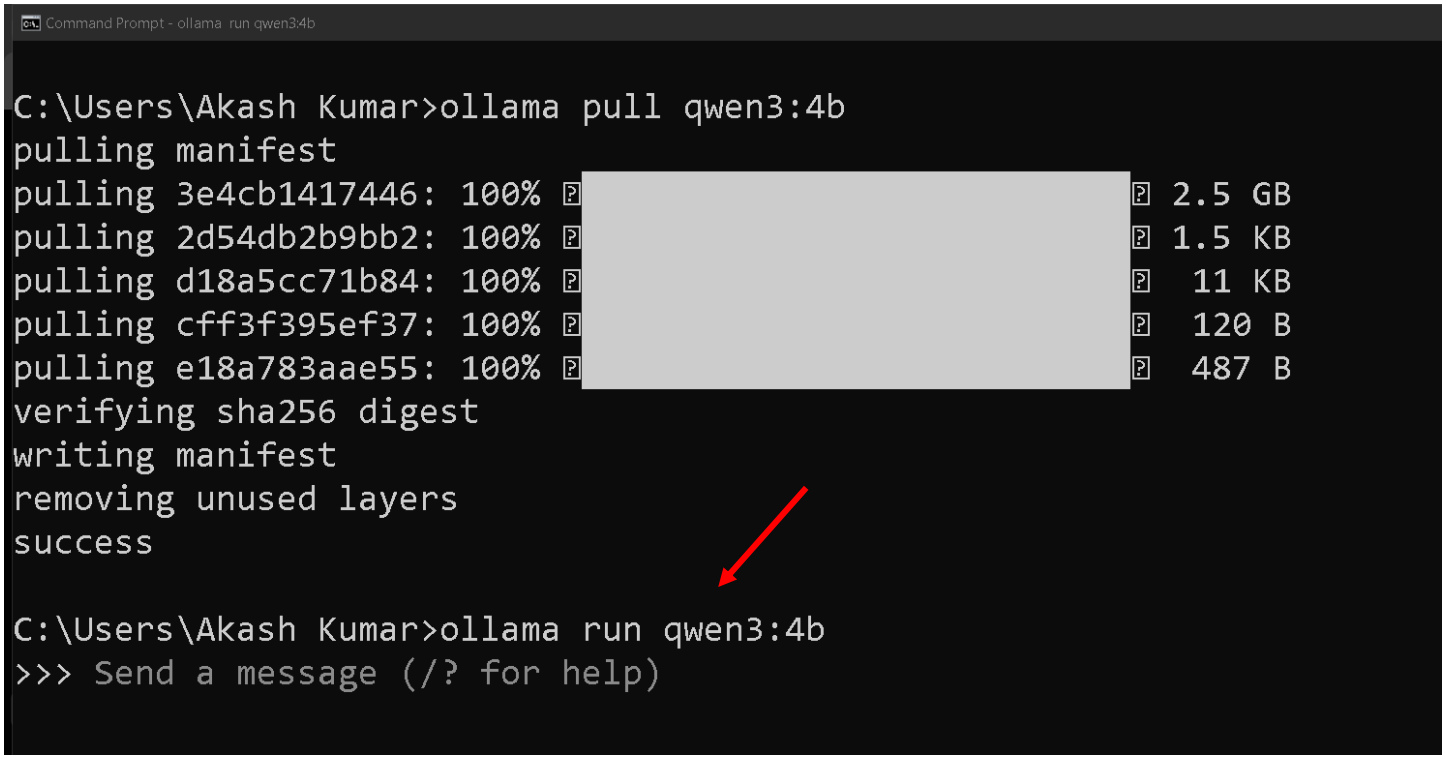
- First download Ollama - Link



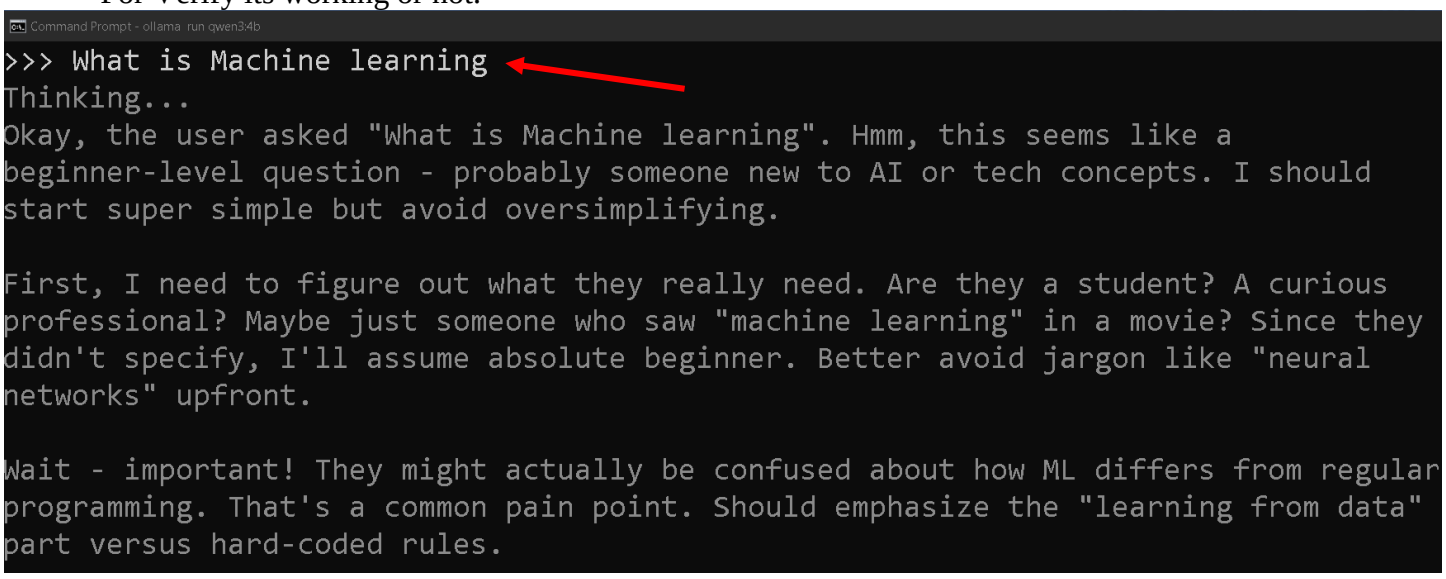
- After downloading Install Ollama.
- Then After installation open command prompt.



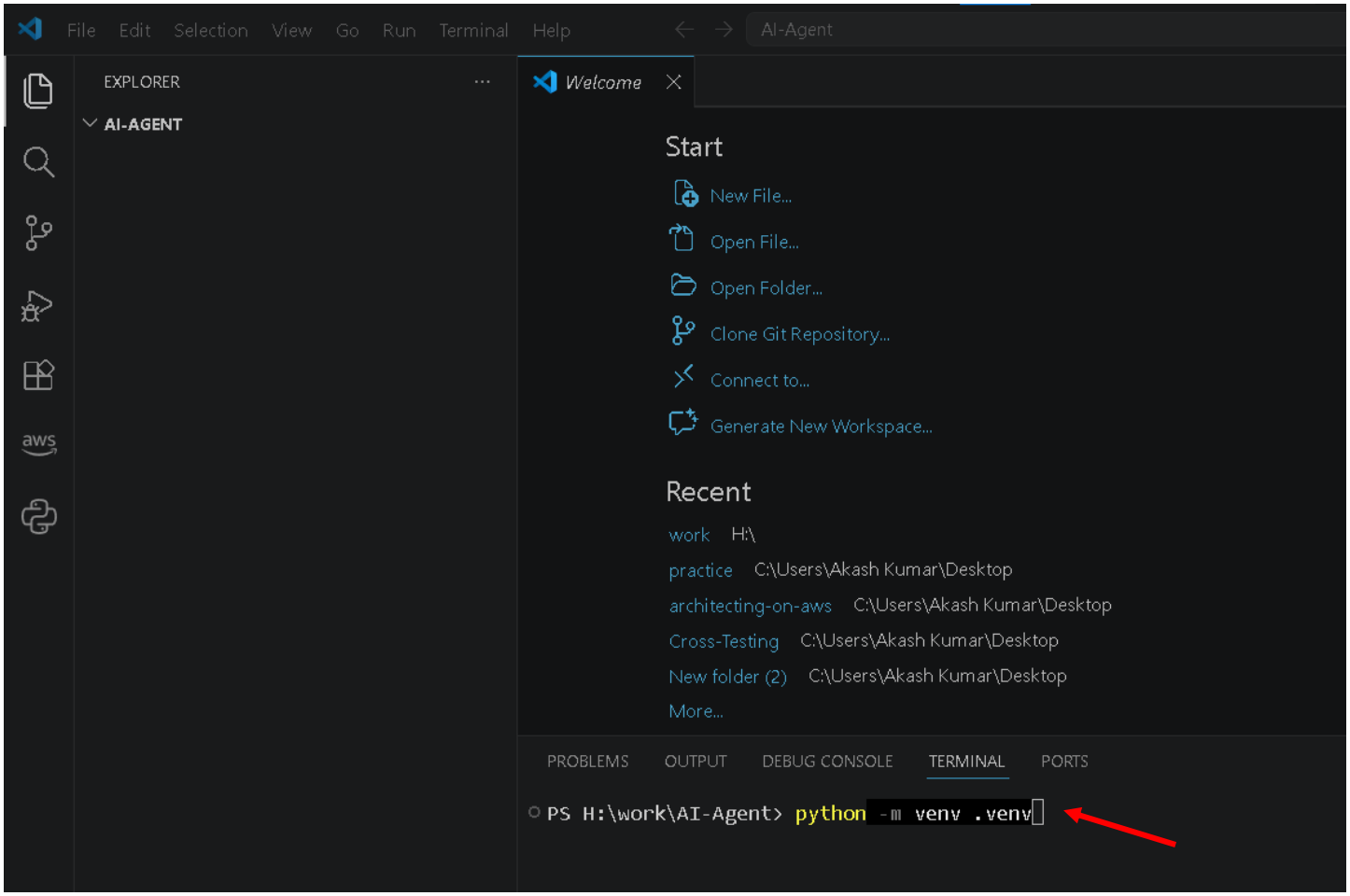
- And Enter ollama pull qwen3:4b
- It will take time.



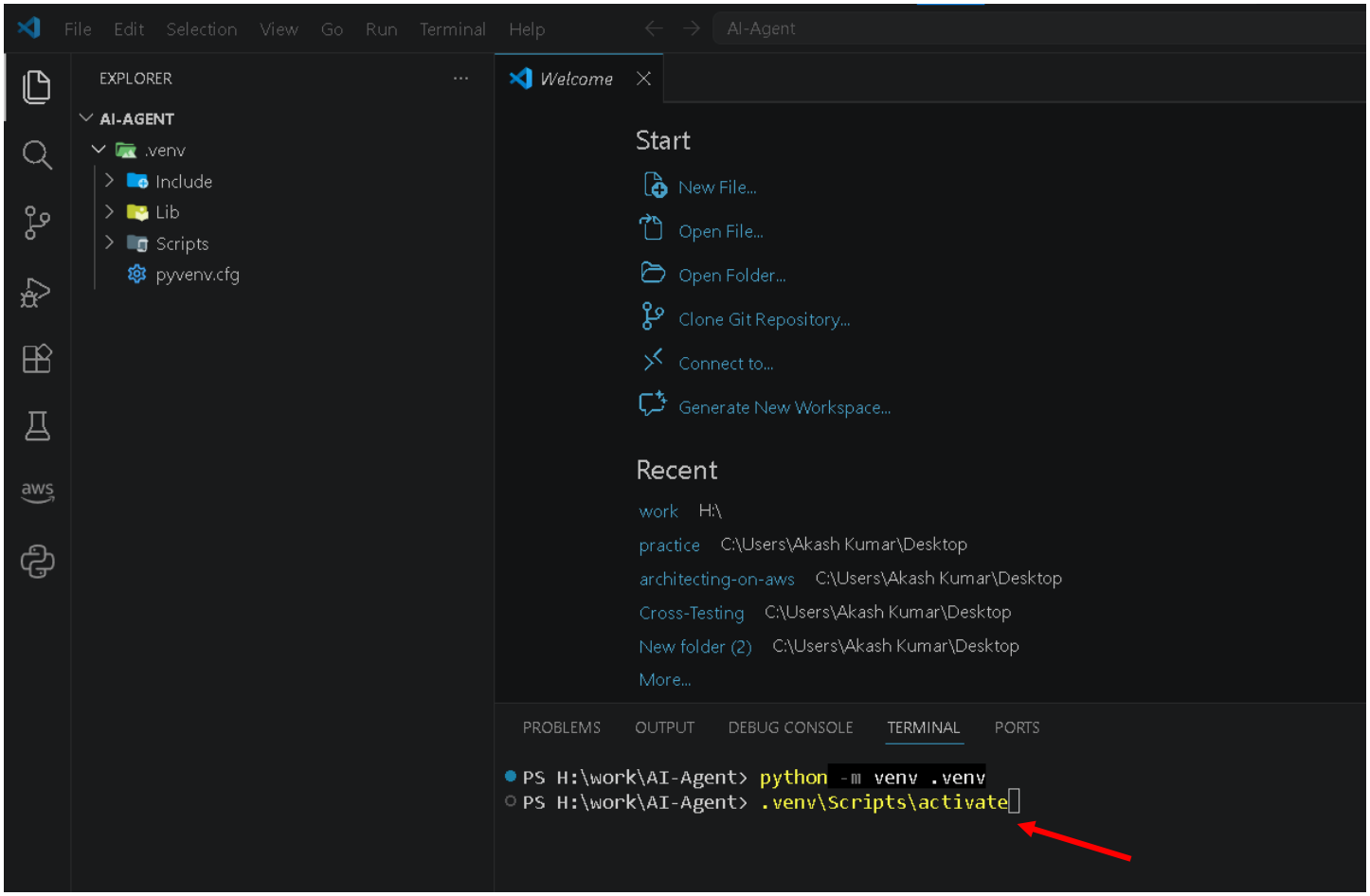
- Now after Downloading AI model.
- Enter ollama run qwen3:4b
- For Verify its working or not.



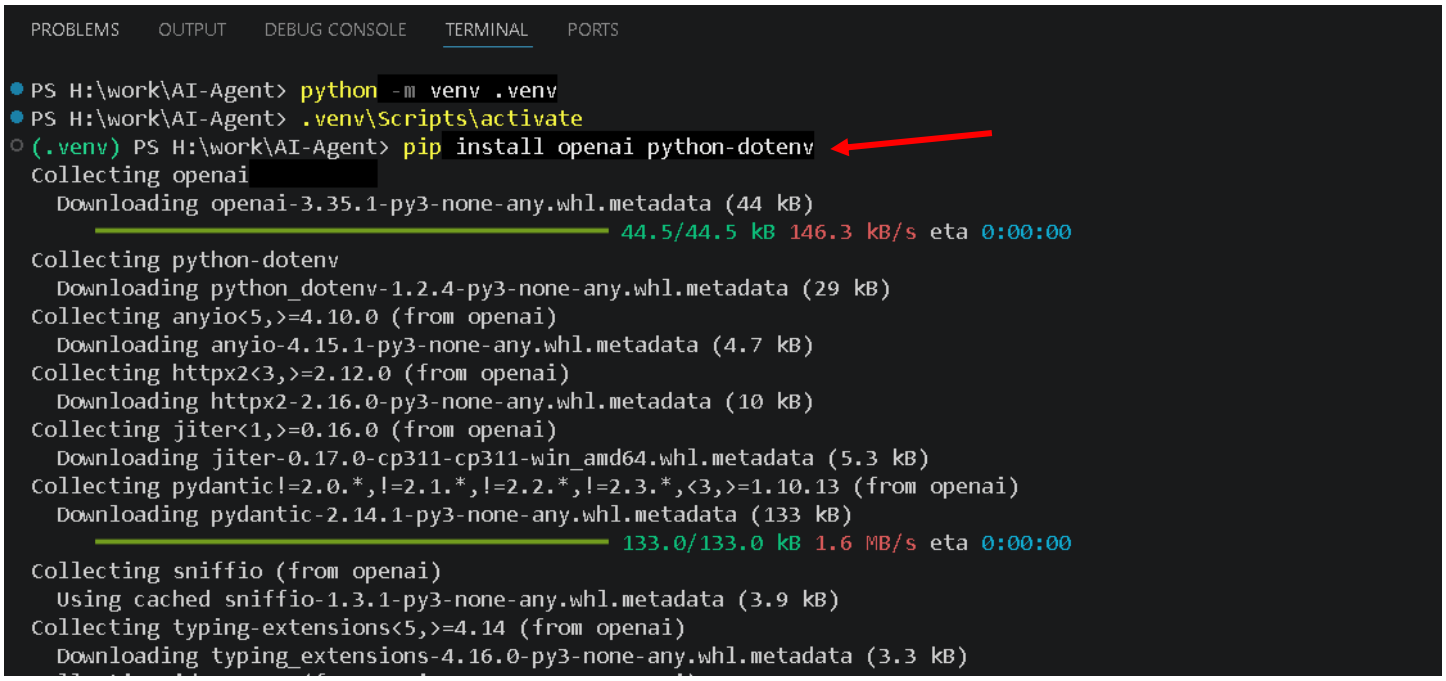
- ask some questions and check its work properly.
- Now create your project folder in your system and open in Vs Code.



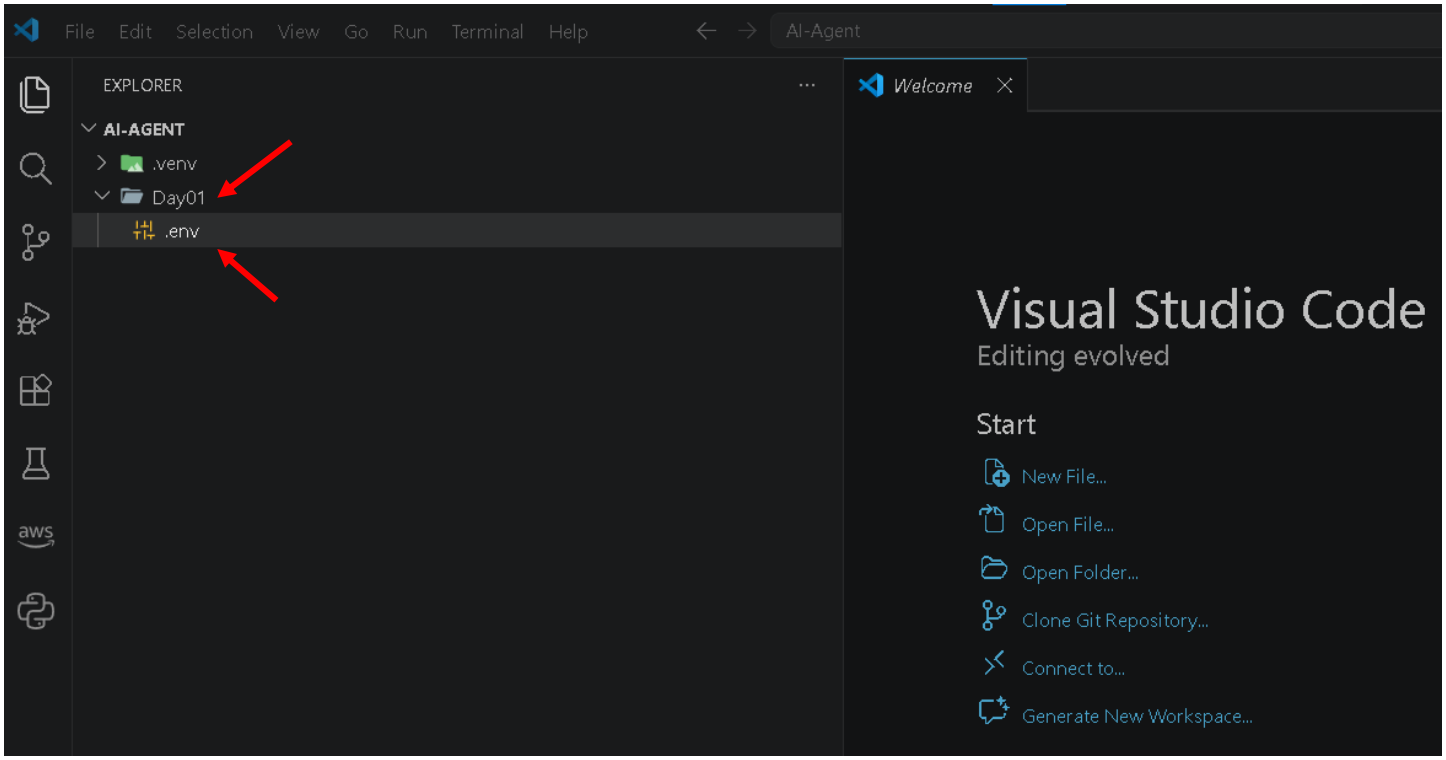
- Our Folder AI-AGENT is opened in Vs Code.
- Now we need to create virtual environment .
- So open terminal and Enter python -m venv .venv



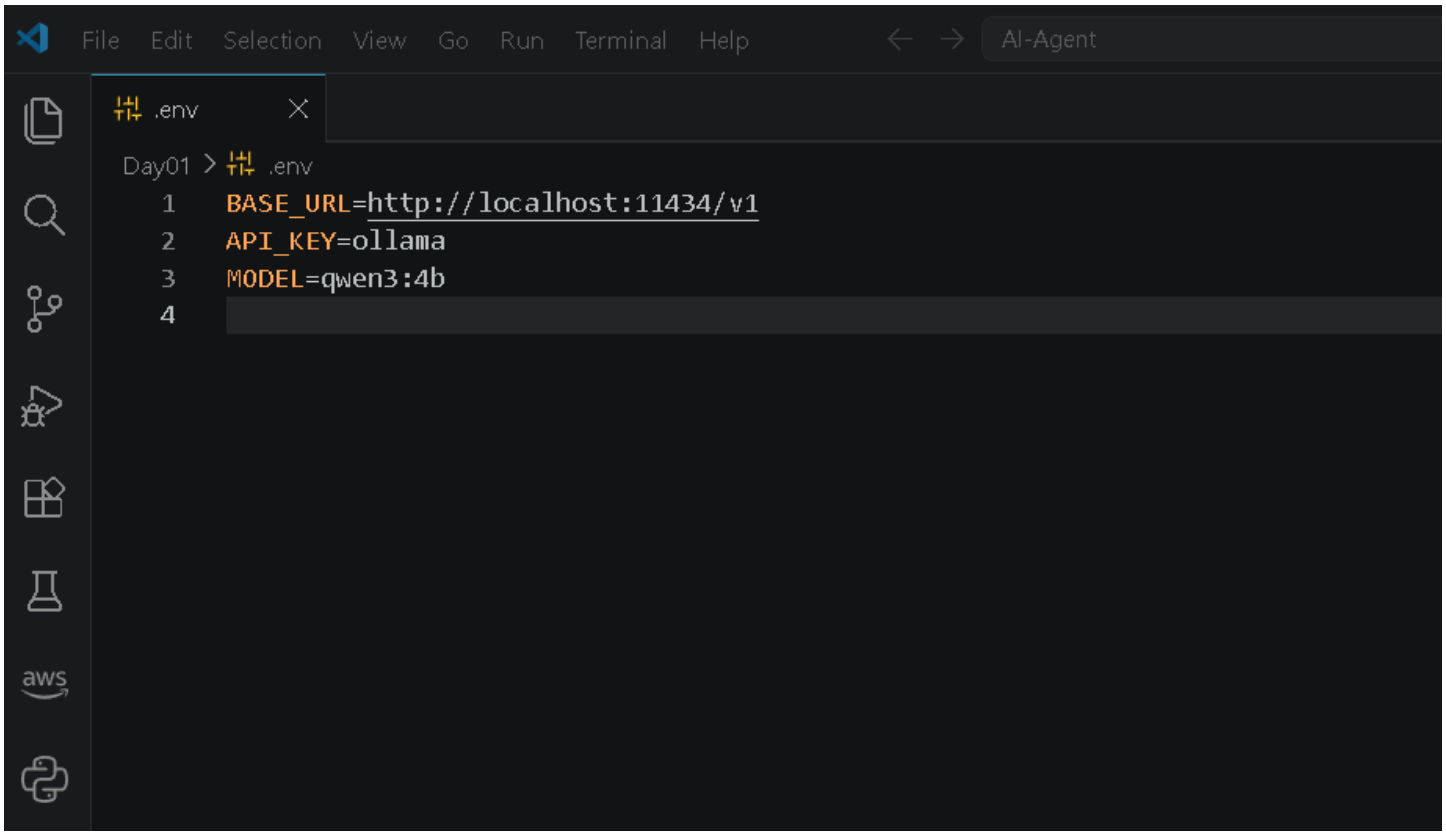
- Now when virtual environment is created we need to activate,
- So enter .venv\Scripts\activate
- Now we need to install some python Library.



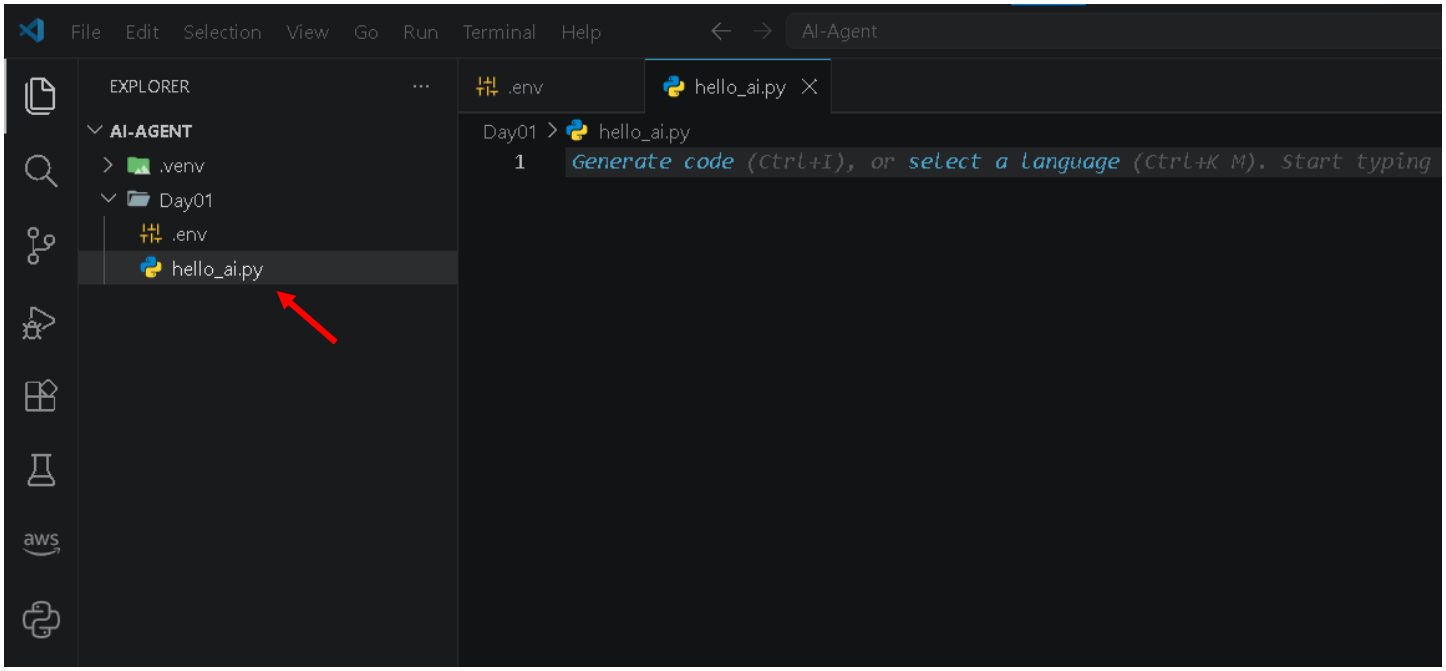
- So We Enter pip install openai python-dotenv



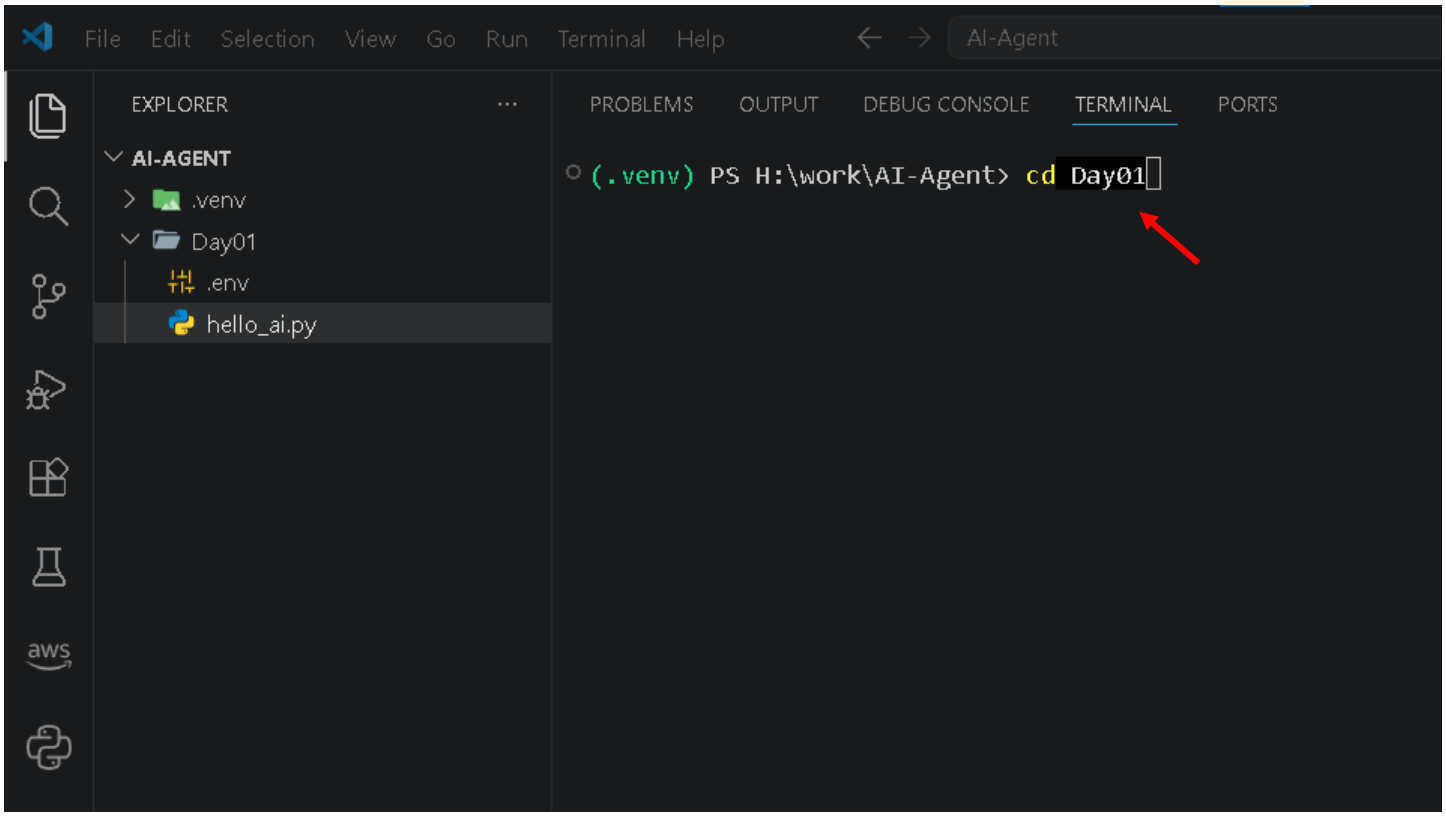
- Now After installation create folder inside AI-AGENT.
- And inside new folder create a new file .env



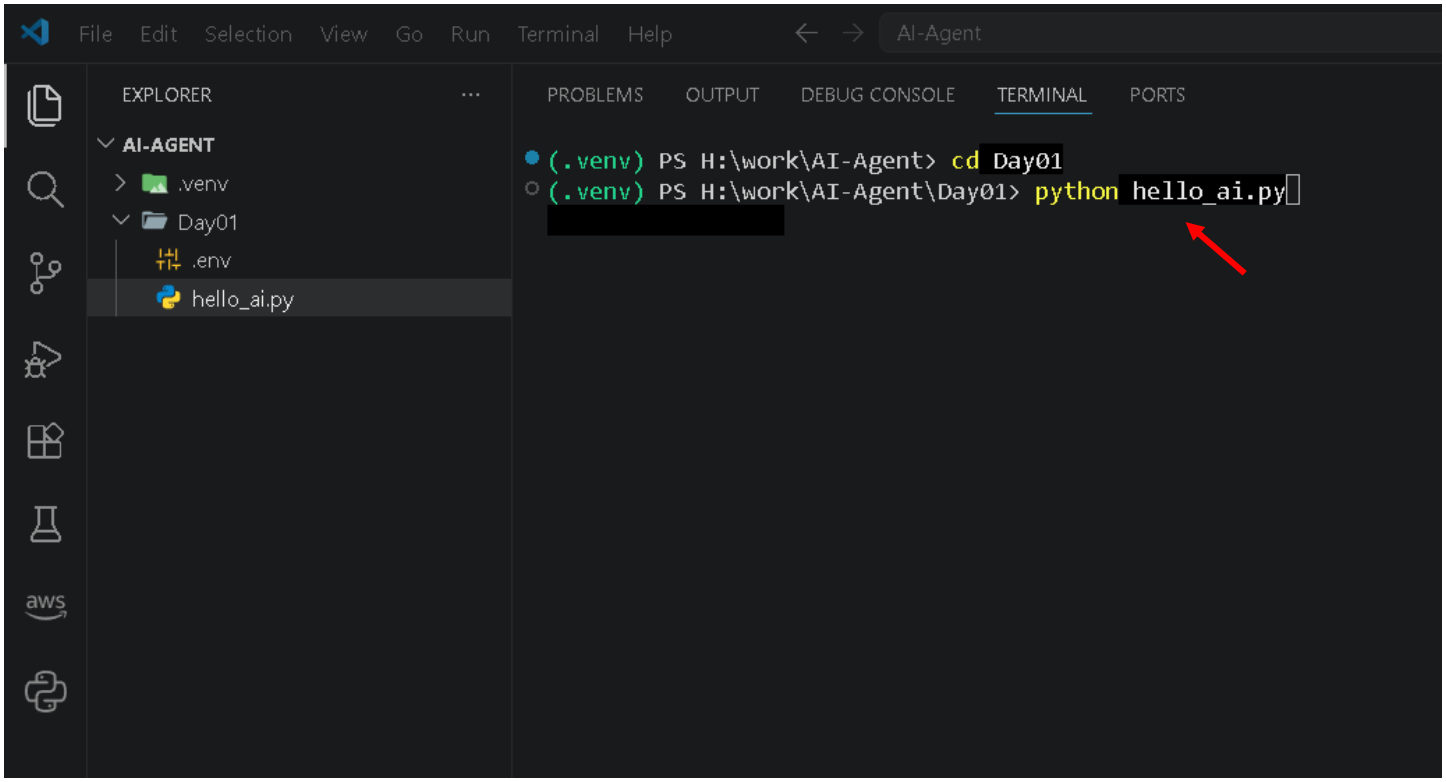
- Now open .env file and Enter these things:-
- BASE_URL=http://localhost:11434/v1
- API_KEY=ollama
- MODEL=qwen3:4b



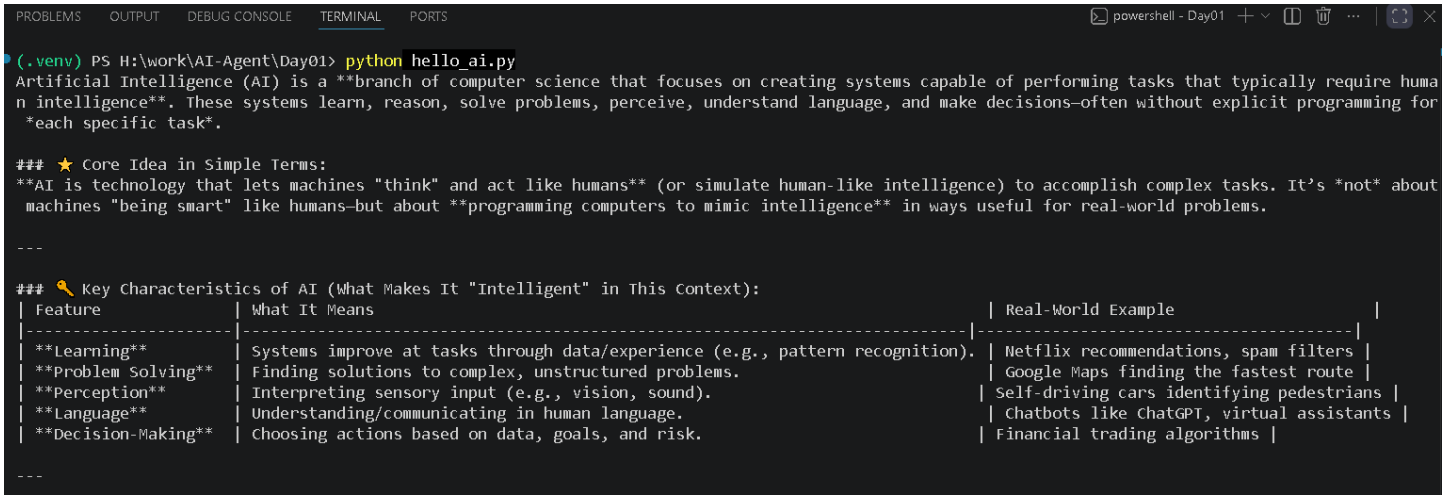
- Now create New file in same folder hello_ai.py
- And Enter the Following codes: -



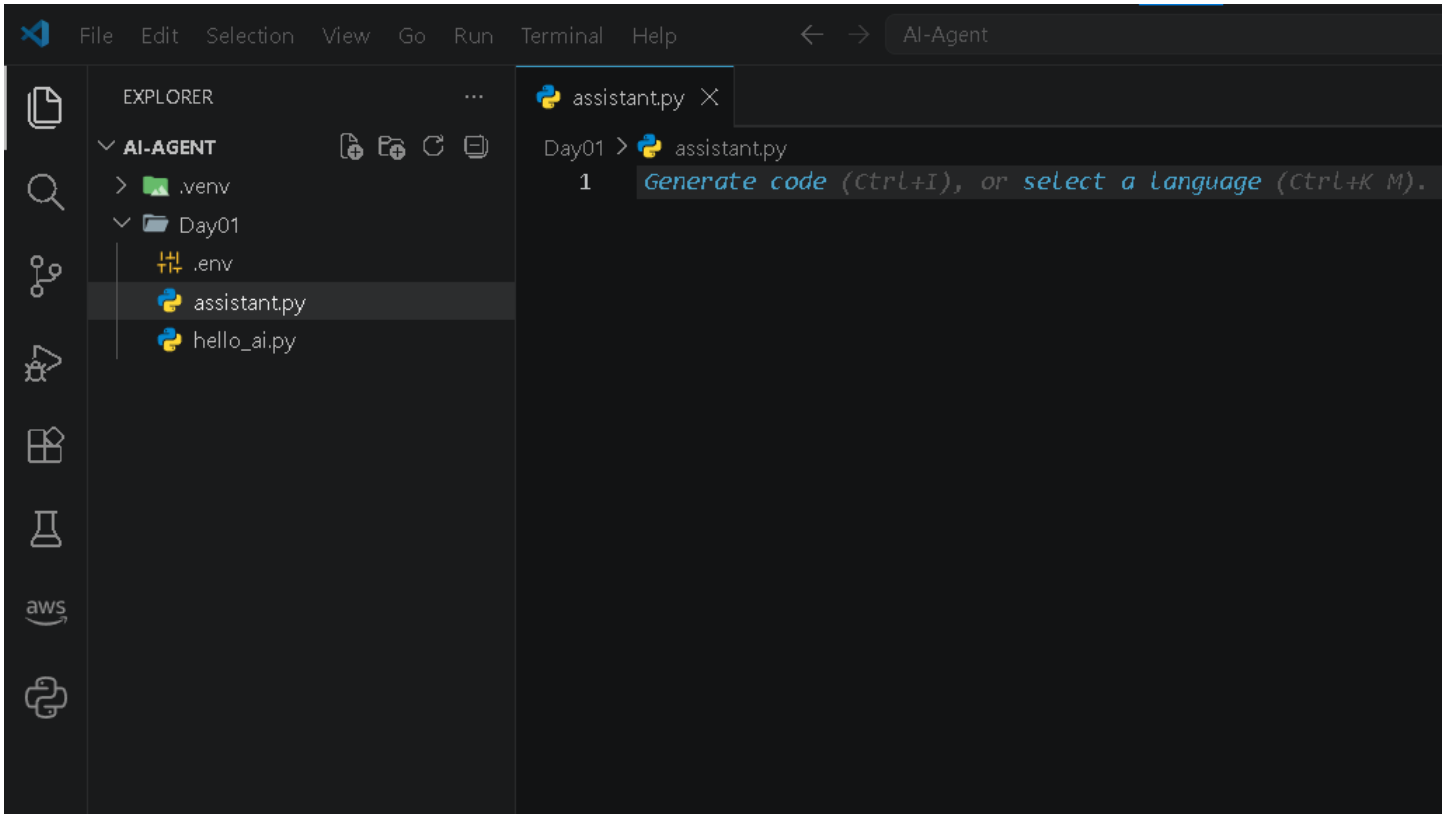
- Now open the terminal and go inside Day01 directory.
- So for this enter cd Day01 in terminal.



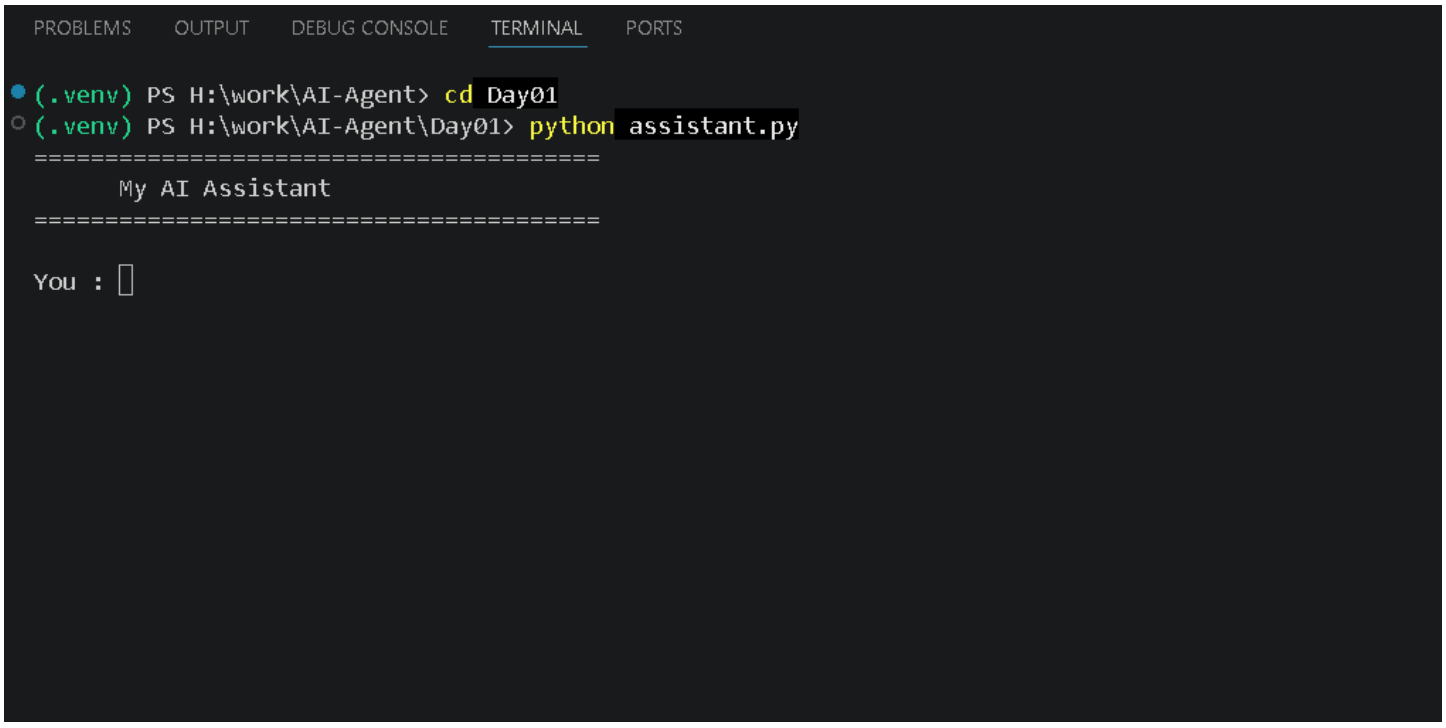
- Now Enter python hello_ai.py



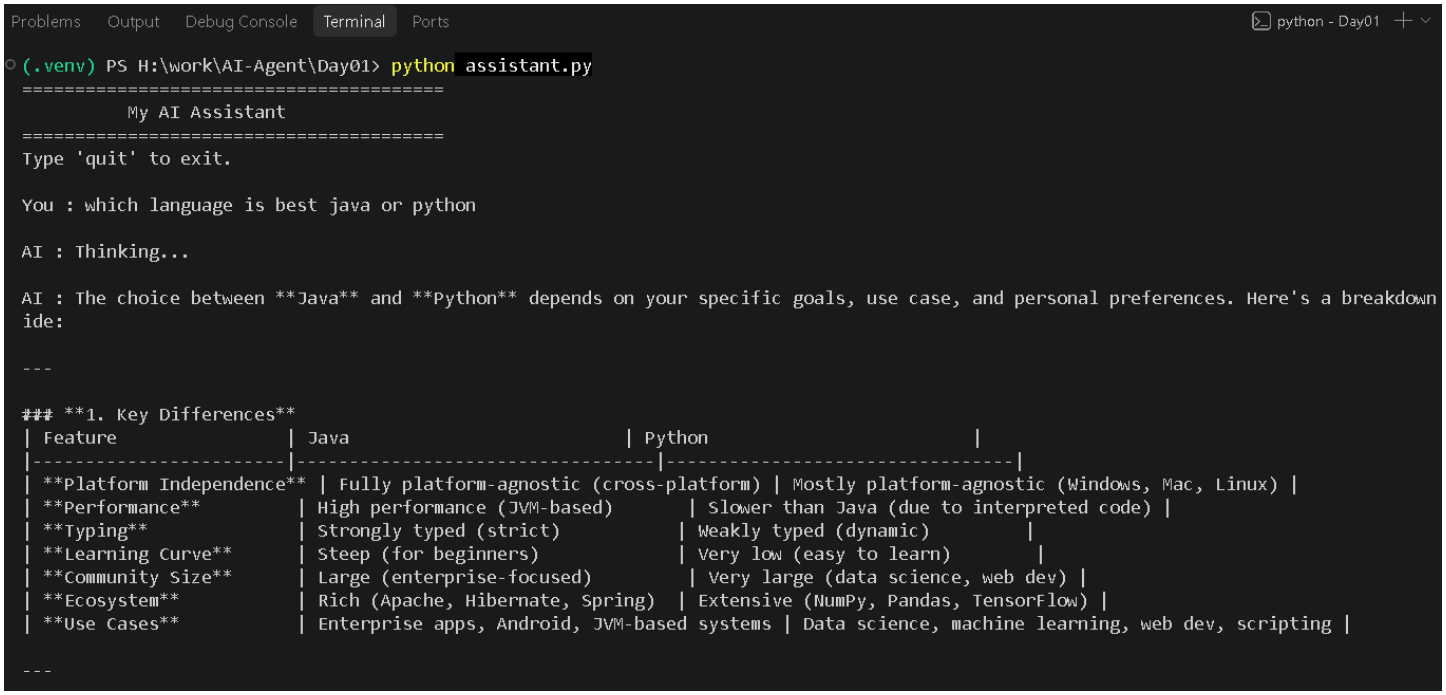
- Now see it will return output according our given content in our code.
- But one issue is that if we need to chat about new topic we need to go inside code and change content.
- So now we can resolve it.



- Now we create a new python file assistant.py
- Now enter the following code: -
- Now open terminal.



- And Enter python assistant.py
- Now see our assistant ask topic in our terminal not in code.
- Now check its work perfectly.



- Now you see I ask which language is best java or python.
- Out chat bot reply perfectly.